

| Question |     |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks available |     |     |       | Maths | Prac |
|----------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total |       |      |
| 4        | (a) |  | <p>Mass defect = <math>((235.01 + 1.01) - (91.90 + 140.89 + 3(1.01)))u</math><br/> [or <math>(235.01 - (91.90 + 140.89 + 2 \times 1.01))] = 0.20 u</math> (1)<br/> Energy released = <math>0.20 \times 931 = 186 \text{ MeV}</math> [190 MeV] (1)<br/> (Accept <math>3.0 \times 10^{-11} \text{ J}</math>) <b>UNIT</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 | 2   |     | 2     | 2     |      |
|          | (b) |  | <p><b>General (G)</b><br/> G1: General shape of curve [correct asymmetry] &amp; axes labelled<br/> G2: Maximum around nucleon number 60 / iron / nickel<br/> G3: Stability linked to BE/nuc to stability<br/> G4: Nucleons react to move towards maximum (on curve)<br/> G5: In doing so there is a loss / reduction in mass<br/> G6: Mass loss linked to energy released = <math>\Delta m c^2</math></p> <p><b>Fusion (Fu)</b><br/> Fu1: Smaller nucleon number nuclei combine<br/> Fu3: Larger nucleons of larger nucleon number formed: energy released, BE/nuc increases, stability increases, lower mass</p> <p><b>Fission (Fi)</b><br/> Fi1: Larger nucleon number split<br/> Fi3: Smaller nucleons of smaller nucleon number formed energy released, BE/nuc increases, stability increases, lower mass</p> <p>Additional point: gradient of curve larger in fusion region than fission resulting in more energy release (per nucleon)</p> <p><b>5-6 marks</b><br/> At least 6 G points (6-8 G) 8 – 11 points<br/> <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i></p> | 6               |     |     | 6     |       |      |